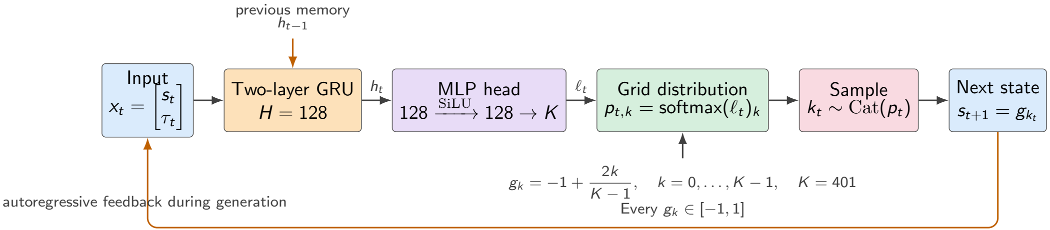
Bounded autoregressive trajectory model



Training at every time step

$$y_t = s_{t+1}^{\text{real}}, \qquad q_{t,i} = 1 - \alpha, \quad q_{t,i+1} = \alpha, \qquad \mathcal{L}_t = -\sum_{k=0}^{K-1} q_{t,k} \log p_{t,k}.$$

$$\underbrace{\text{real } s_t \text{ as input}}_{\text{teacher forcing}} \longrightarrow \underbrace{\text{two-bin interpolated target } q_t}_{\text{continuous data on a discrete grid}} \longrightarrow \underbrace{\text{categorical cross-entropy}}_{\text{maximize next-state likelihood}}.$$